QA-TEST QA-E2E-report-I 1: simulation HeatingAbsorptionCoolingPlant

1 Summary

Cost of thermal supply	Investment	Operating cost	CO2 emissions
49.28 EUR/MWh	23.96 MEUR	4.978 MEUR/yr	20.96 kt

Heat delivered	Cold delivered	Geothermal and heat-pump share
111.7 GWh	23.74 GWh	39 %

Figure 1.1 - Key figures of the design

The design delivers 111.7 GWh of heat and 23.74 GWh of cold a year. The levelised cost is 49.28 EUR/MWh. Geothermal, with any heat-pump electricity, covers 39 % of the heating demand. The largest rated units are Absorption cooler 40 MW, Peak boiler 30 MW, Absorption chiller 10 MW. It emits 20.96 kt of CO2 over the run. All 15 assessed checks pass.

2 Case and demand

The design runs on case hackathon_2026, version 3. The simulation runs from 2027-01-01 to 2028-01-01, in steps of 1 d.

Quantity	Value	Uncertainty
Air pressure	101,300 Pa	-
Air temperature	10 degC	-
Fluid density	1,030 kg/m3	-
Fluid viscosity	0.4 mPa.s	-
Fluid dissolved gas CH4	0.8	-
Fluid dissolved gas CO2	0.2	-
Fluid gas liquid ratio	0.8 Nm3/m3	-
Fluid sand concentration	20 mg/l	-
Ground gradient	30 K/km	normal, 25 to 35 K/km
Ground surface temperature	12 degC	-
Reservoir pressure	250 barg	-
Reservoir thickness	100 m	-
Reservoir permeability	200 mD	triangular, 100 to 400 mD
Reservoir reference depth	2,500 m	-

Table 2.1 - Site, reservoir and fluid values of the case

The reservoir has a depth of 2,500 m, a thickness of 100 m and a permeability of 200 mD. 2 of the 14 values carry an uncertainty range, which the studies of section 7 can vary.

Service	Supply	Return	Annual demand	Peak (daily mean)	Met
Cooling	20 degC	25 degC	23.74 GWh	10 MW	100 %
Heating	100 degC	60 degC	111.7 GWh	30 MW	100 %

Table 2.2 - Demand of each service and how much of it was met

Cooling demand totals 23.74 GWh with a peak daily mean of 10 MW on 2027-07-19 and a mean of 2.71 MW. Heating demand totals 111.7 GWh with a peak daily mean of 30 MW on 2027-01-28 and a mean of 12.75 MW. The demand profiles come from data/demand.csv (cooling_demand_MWcool) and data/demand.csv (heating_demand_MWth).

3 Plant design

Design simulation HeatingAbsorptionCoolingPlant from a study of kind Run the design, plant type Heating absorption cooling plant.

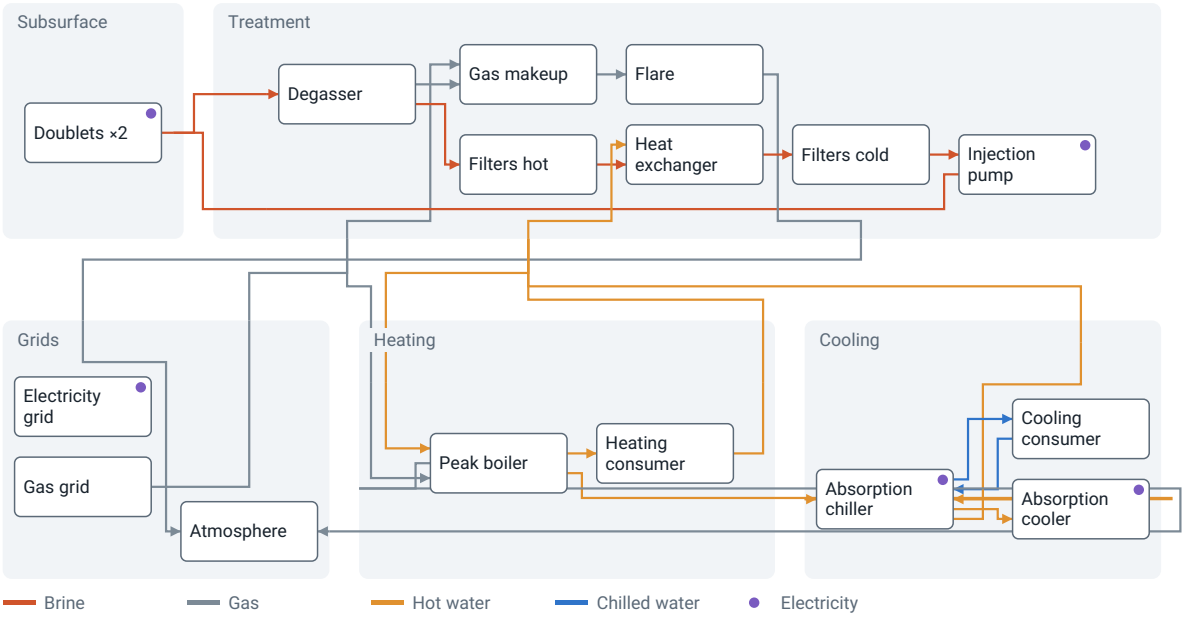


Figure 3.1 - Process flowsheet of the plant

The flowsheet shows the 25 modules of the plant and the 48 connections between them, carrying air, brine, chilled_water, electricity, gas, heat and hot_water. The costed units fall into 9 subsurface, 3 treatment and gas, 2 heating, 2 cooling and 4 grids units; subsections 3.1 to 3.5 give each group with its size and use.

3.1 Subsurface

Module	Kind	Rated	Peak	Used	Capital cost
d0	Production pump	300 m3/h	300 m3/h	100 %	0.2277 MEUR
d1	Production pump	300 m3/h	300 m3/h	100 %	0.2277 MEUR
D0 well	Drilling	-	-	-	2.659 MEUR
D1 well	Drilling	-	-	-	2.659 MEUR
D0 riser	Drilling	-	-	-	0.4392 MEUR
D1 riser	Drilling	-	-	-	0.4392 MEUR
D0 injector	Drilling	-	-	-	3.098 MEUR
D1 injector	Drilling	-	-	-	3.098 MEUR
Injection pump	Injection pump	-	-	-	0.1874 MEUR

Table 3.1 - Subsurface units: rating, use and capital cost

9 units make up the subsurface part, with a capital cost of 13.03 MEUR, 54 % of the plant's total. The largest is D0 injector. d0 and d1 ran at their full rating, so they limit what the plant can deliver.

3.2 Treatment and gas

Module	Kind	Rated	Peak	Used	Capital cost
Flare	Flare	-	-	-	0.1 MEUR
Degasser	Degasser	-	-	-	1 MEUR
Gas makeup	Gas makeup	-	-	-	0.05 MEUR

Table 3.2 - Treatment and gas units: rating, use and capital cost

3 units make up the treatment and gas part, with a capital cost of 1.15 MEUR, 4.8 % of the plant's total. The largest is Degasser.

3.3 Heating

Module	Kind	Rated	Peak	Used	Capital cost
Peak boiler	Boiler	30 MW	17.64 MW	59 %	1.822 MEUR
Heat exchanger	Heat exchanger	1.5 MW/K	-	-	0.75 MEUR

Table 3.3 - Heating units: rating, use and capital cost

2 units make up the heating part, with a capital cost of 2.572 MEUR, 11 % of the plant's total. The largest is Peak boiler, rated 30 MW, which ran at up to 59 % of its rating.

3.4 Cooling

Module	Kind	Rated	Peak	Used	Capital cost
Absorption cooler	Cooler	40 MW	24.39 MW	61 %	4 MEUR
Absorption chiller	Absorption chiller	10 MW	10 MW	100 %	3 MEUR

Table 3.4 - Cooling units: rating, use and capital cost

2 units make up the cooling part, with a capital cost of 7 MEUR, 29 % of the plant's total. The largest is Absorption cooler, rated 40 MW, which ran at up to 61 % of its rating. Absorption chiller ran at its full rating, so it limits what the plant can deliver.

3.5 Grids

Module	Kind	Rated	Peak	Used	Capital cost
Cooling return pipe	Pipe	100 m	-	-	0.05 MEUR
Cooling supply pipe	Pipe	100 m	-	-	0.05 MEUR
Heating return pipe	Pipe	100 m	-	-	0.05 MEUR

Module	Kind	Rated	Peak	Used	Capital cost
Heating supply pipe	Pipe	100 m	-	–	0.05 MEUR

Table 3.5 – Grids units: rating, use and capital cost

4 units make up the grids part, with a capital cost of 0.2 MEUR, 0.83 % of the plant's total. The largest is Cooling return pipe, rated 100 m.

4 Performance

Figure	Value
Gas bought	65.19 GWh
Heat from the boilers	76.83 GWh
Cold delivered	23.74 GWh
Heat delivered	111.7 GWh
Electricity bought	9.134 GWh
Fuel burned on site	85.37 GWh
Produced gas burned (fuel burned less gas bought)	20.18 GWh

Table 4.1 - Energy balance of the run

The plant delivers 111.7 GWh of heat against a demand of 111.7 GWh, so it meets 100 % of it. The boilers give 76.83 GWh, 69 % of the heat delivered. The plant burns 85.37 GWh of fuel on site: 65.19 GWh of gas bought and 20.18 GWh of gas produced with the brine. It buys 9.134 GWh of electricity and 65.19 GWh of gas. Geothermal, with any heat-pump electricity, covers 39 % of the heating demand.



Figure 4.1 - Source that covered the demand on each day

Of the 365 cooling days, 365 had a demand. Of the days with a demand, 365 were covered by Plant alone and 0 fell short. Of the 365 heating days, 365 had a demand. Of the days with a demand, 0 were covered by Geothermal alone, 365 also needed Backup and 0 fell short.

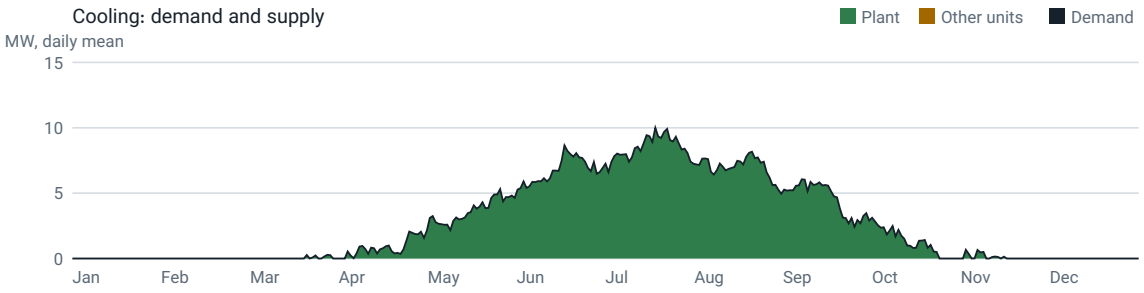


Figure 4.2 - Cooling: daily mean demand and supply

Cooling demand peaks at 10 MW (daily mean) on 2027-07-19 and averages 2.71 MW. Plant gives the whole supply. 2.01e-12 MWh of demand is not met.

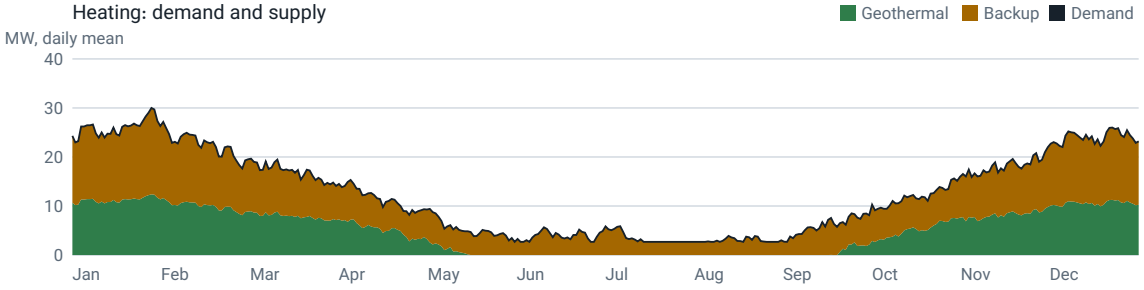


Figure 4.3 - Heating: daily mean demand and supply

Heating demand peaks at 30 MW (daily mean) on 2027-01-28 and averages 12.75 MW.

Backup runs on 365 days, at most 17.64 MW on 2027-01-28. 1.137e-11 MWh of demand is not met.

Service	Main supply	Days	Other supply	Days	Short	No demand
Cooling	Plant	365	-	-	0	0
Heating	Geothermal	0	Backup	365	0	0

Table 4.2 - Days of each service by the units that covered them

5 Economics

From case version 3, plant version 2.

Figure	Value
Cost of cold	56.21 EUR/MWh
Cost of heat	47.81 EUR/MWh
Cost of thermal supply	49.28 EUR/MWh
Operating cost	4.978 MEUR/yr
Investment	23.96 MEUR

Table 5.1 - Cost figures of the design

The levelised cost of thermal supply is 49.28 EUR/MWh, from an annualised cost of 6.678 MEUR/yr. The plant costs 23.96 MEUR to build and 4.978 MEUR/yr to run.

Capital cost	Amount	Share
Wells	12.85 MEUR	54 %
Surface plant	10.76 MEUR	45 %
Other	0.35 MEUR	1.5 %

Table 5.2 - Capital cost by part of the plant

Wells take the largest share of the capital cost, 54 % (12.85 MEUR); the table splits the rest.

Item	Amount	Unit cost	Source
Drilling	12.39 MEUR	1,239 EUR/m	economics.json, capex.drilling
Cooler	4 MEUR	0.1 EUR/W	economics.json, capex.cooler
Absorption chiller	3 MEUR	0.3 EUR/W	economics.json, capex.absorption_chiller
Boiler	1.822 MEUR	0.06073 EUR/W	economics.json, capex.boiler
Degasser	1 MEUR	1,000,000 EUR	economics.json, capex.degasser
Heat exchanger	0.75 MEUR	0.5 EUR/(W/K)	economics.json, capex.heat_exchanger
Production pump	0.4553 MEUR	0.5 EUR/W	economics.json, capex.production_pump
Pipe	0.2 MEUR	500 EUR/m	economics.json, capex.pipe
Injection pump	0.1874 MEUR	0.5 EUR/W	economics.json, capex.injection_pump
Flare	0.1 MEUR	100,000 EUR	economics.json, capex.flare

Item	Amount	Unit cost	Source
Gas makeup	0.05 MEUR	50,000 EUR	economics.json, capex.gas_makeup

Table 5.3 – Capital cost by kind of unit, with its unit cost

The largest items are drilling (52 %), cooler (17 %) and absorption chiller (13 %); together they make 81 % of the capital cost.

Operating cost	Amount	Share
Fuel	2.604 MEUR/yr	52 %
CO2 price	1.04 MEUR/yr	21 %
Electricity	0.9666 MEUR/yr	19 %
Fixed	0.3675 MEUR/yr	7.4 %
Waste	0 MEUR/yr	0 %
Chemical	0 MEUR/yr	0 %

Table 5.4 – Yearly operating cost by component

Fuel is the largest operating cost, 2.604 MEUR/yr or 52 % of the total. The run has no waste and chemical cost.

6 Emissions

Source	CO2	Share
Fuel burnt in the plant	18.22 kt	87 %
Methane released	0 kt	0 %
Grid electricity bought	2.74 kt	13 %
Total	20.96 kt	100 %

Table 6.1 - CO2 emissions of the run by source

The plant emits 20.96 kt of CO2 over the run, 154.8 kg/MWh of energy delivered. Fuel burnt in the plant is the largest source, 87 % of the total. They come from 85.37 GWh of fuel and 9.134 GWh of grid electricity.

7 Studies

The result comes from the run the design study simulation HeatingAbsorptionCoolingPlant.

This result is not a sensitivity or uncertainty study.

8 Checks

15 passed, 0 failed, 0 not assessed.

Check	Status	Value	Limit	Margin	Worst day
Production flow range (d0)	pass	300 m3/h	50 to 300 m3/h	0 m3/h	2027-01-01
Production flow range (d1)	pass	300 m3/h	50 to 300 m3/h	0 m3/h	2027-01-27
Brine cooling limit (d0)	pass	20.7 K	≤ 40 K	19.3 K	2027-04-24
Brine cooling limit (d1)	pass	20.73 K	≤ 40 K	19.27 K	2027-05-08
Running time limit (d0)	pass	330 d	≤ 330 d	0 d	–
Running time limit (d1)	pass	330 d	≤ 330 d	0 d	–
Pump efficiency (d0)	pass	70 %	= 70 %	–	–
Pump efficiency (d1)	pass	70 %	= 70 %	–	–
Full reinjection (d0)	pass	1	1 to 1	0	2027-01-02
Full reinjection (d1)	pass	1	1 to 1	0	2027-01-20
Doublet count	pass	2	1 to 3	1	–
Heat delivered (Heating consumer)	pass	0 %	≤ 1 %	1 %	–
Cold delivered (Cooling consumer)	pass	0 %	≤ 1 %	1 %	–
Heating supply temperature (Heating consumer)	pass	100 degC	≥ 100 degC	0 degC	2027-01-01
Cooling supply temperature (Cooling consumer)	pass	20 degC	≤ 20 degC	0 degC	2027-03-22

Table 8.1 - Constraint checks of the design

No check fails. Every check could be assessed.

The checks are the 15 rules of the case. They bound brine cooling, count, production flow, pump efficiency, reinjection ratio, temperature and unmet share on the cooling supply, doublet and heating supply. The engine can also bound approach, co2, cold output, cop, duty, electricity bought, fuel input, gas bought, gas burned, heat output, heat surplus, injection temperature, intake gas fraction, intake pressure, lift, power output, production temperature, pump power, unmet and wellhead pressure; the case sets no rule on them, so they are not checked. Every check judges the run of 365 d only: nothing that develops over a longer life of the plant is checked.

9 Assumptions and sources

Input	Value	Source
Case	hackathon_2026, version 3	Project case
Plant	version 2	Project plant
Economics	economics.json	Case file
Simulation period	2027-01-01 to 2028-01-01, in steps of 1 d	Case

Table 9.1 - Inputs the result was computed from

Every figure of this report is computed by the engine from these inputs; a value the inputs do not give is reported as missing, never assumed.

Economic value	Amount	Source
Currency	EUR	economics.json
Price year	2,015	economics.json
Discount rate	0.05	economics.json
Analysis period	25 yr	economics.json
Capex drilling	1,239 EUR/m	economics.json
Capex heat pump	0.7379 EUR/W	economics.json
Capex boiler	0.06073 EUR/W	economics.json
Capex chp	0.9616 EUR/W	economics.json
Capex degasser	1,000,000 EUR	economics.json
Capex compression chiller	0.3 EUR/W	economics.json
Capex production pump	0.5 EUR/W	economics.json
Capex injection pump	0.5 EUR/W	economics.json
Capex absorption chiller	0.3 EUR/W	economics.json
Capex adsorption chiller	0.3 EUR/W	economics.json
Capex heat exchanger	0.5 EUR/(W/K)	economics.json
Capex flare	100,000 EUR	economics.json
Capex gas makeup	50,000 EUR	economics.json
Capex cooler	0.1 EUR/W	economics.json
Capex pipe	500 EUR/m	economics.json
Opex fixed drilling	1 %/yr	economics.json
Opex fixed heat pump	0.2239 %/yr	economics.json
Opex fixed boiler	3.25 %/yr	economics.json
Opex fixed chp	1.026 %/yr	economics.json
Opex fixed degasser	2 %/yr	economics.json
Opex fixed compression chiller	2 %/yr	economics.json
Opex fixed production pump	2 %/yr	economics.json
Opex fixed injection pump	2 %/yr	economics.json
Opex fixed absorption chiller	2 %/yr	economics.json

Economic value	Amount	Source
Opex fixed adsorption chiller	2 %/yr	economics.json
Opex fixed heat exchanger	1 %/yr	economics.json
Opex fixed flare	2 %/yr	economics.json
Opex fixed gas makeup	2 %/yr	economics.json
Opex fixed cooler	2 %/yr	economics.json
Opex fixed pipe	0.5 %/yr	economics.json
Opex variable chemical	3 EUR/kg	economics.json
Opex variable waste disposal	0.2 EUR/kg	economics.json
Energy electricity	105.8 EUR/MWh	economics.json
Energy gas	39.92 EUR/MWh	economics.json
Carbon price	57.03 EUR/t	economics.json
Revenue heat	60 EUR/MWh	economics.json
Revenue cold	60 EUR/MWh	economics.json
Accounting allocation heat cost share	0.8	economics.json
Accounting gwp standard	IPCC AR6 WG1 table 7.15, GWP100 of fossil methane	economics.json
Accounting gwp CH4	29.8	economics.json

Table 9.2 - Economic values of the case

The case states 44 economic values. 4 of them carry an uncertainty range.

Appendix A – Agent log and tokens

No assistant run was chosen.

Appendix B – Modules and connections

Module	Kind	Size	Peak	Used
Cooling return pipe	Pipe	100 m	-	-
Cooling supply pipe	Pipe	100 m	-	-
Heating return pipe	Pipe	100 m	-	-
Heating supply pipe	Pipe	100 m	-	-

Plant: HeatingAbsorptionCoolingPlant

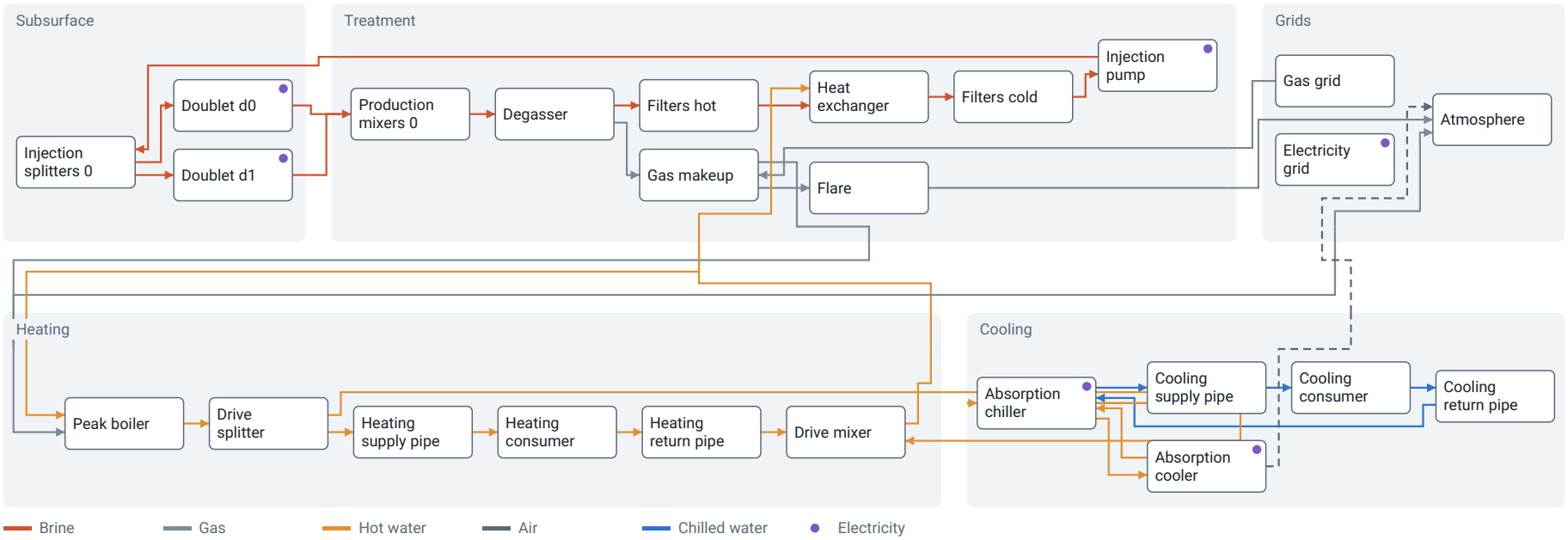
Module	Type	Category	Connected to	Parameters
d0	Doublet	plants	return_in <- injection_splitters.0.branch1_out, power_in <- electricity_grid.power_out	85
d1	Doublet	plants	return_in <- injection_splitters.0.branch2_out, power_in <- electricity_grid.power_out	85
production_mixers.0	Mixer	piping	branch1_in <- d0.supply_out, branch2_in <- d1.supply_out	0
injection_splitters.0	Splitter	piping	fluid_in <- injection_pump.fluid_out	0
degasser	Degasser	treatment	fluid_in <- production_mixers.0.fluid_out	4
filters.hot	Filter	treatment	fluid_in <- degasser.liquid_out	3
filters.cold	Filter	treatment	fluid_in <- heat_exchanger.hot_out	3

Module	Type	Category	Connected to	Parameters
heat_exchanger	HeatExchanger	exchangers	hot_in <- filters.hot.fluid_out, cold_in <- drive_mixer.fluid_out	8
injection_pump	InjectionPump	pumps	fluid_in <- filters.cold.fluid_out, power_in <- electricity_grid.power_out	7
heating_return_pipe	Pipe	piping	fluid_in <- heating_consumer.return_out, air_in <- atmosphere.air_out	6
peak_boiler	Boiler	heating	fuel_in <- gas_makeup.fuel_out, fluid_in <- heat_exchanger.cold_out	3
absorption_chiller	AbsorptionUnit	cooling	generator_in <- drive_splitter.branch2_out, chilled_in <- cooling_return_pipe.fluid_out, power_in <- electricity_grid.power_out, cooling_in <- absorption_cooler.fluid_out	10
drive_splitter	Splitter	piping	fluid_in <- peak_boiler.fluid_out	0
drive_mixer	Mixer	piping	branch1_in <- heating_return_pipe.fluid_out, branch2_in <- absorption_chiller.generator_out	0

Module	Type	Category	Connected to	Parameters
heating_supply_pipe	Pipe	piping	air_in <- atmosphere.air_out, fluid_in <- drive_splitter.branch1_out	6
heating_consumer	HeatingGrid	grids	supply_in <- heating_supply_pipe.fluid_out	4
cooling_return_pipe	Pipe	piping	fluid_in <- cooling_consumer.return_out, air_in <- atmosphere.air_out	6
cooling_supply_pipe	Pipe	piping	air_in <- atmosphere.air_out, fluid_in <- absorption_chiller.chilled_out	6
cooling_consumer	CoolingGrid	grids	supply_in <- cooling_supply_pipe.fluid_out	5
gas_grid	GasGrid	grids		1
electricity_grid	ElectricityGrid	grids		2
gas_makeup	GasMakeup	piping	gas_in <- degasser.gas_out, grid_in <- gas_grid.gas_out	0
flare	Flare	treatment	gas_in <- gas_makeup.excess_out	1

Module	Type	Category	Connected to	Parameters
atmosphere	Atmosphere	environment	gas_in <- peak_boiler.flue_out, gas_in <- flare.flue_out, heat_in <- flare.loss_out, air_in <- absorption_cooler.air_out, heat_in <- heating_return_pipe.loss_out, heat_in <- heating_supply_pipe.loss_out, heat_in <- cooling_return_pipe.loss_out, heat_in <- cooling_supply_pipe.loss_out, heat_in <- peak_boiler.loss_out	1
absorption_cooler	Cooler	exchangers	fluid_in <- absorption_chiller.cooling_out, air_in <- atmosphere.air_out, power_in <- electricity_grid.power_out	6

Total parameters: 256


Subsurface
Injection splitters 0
Doublet d0
Doublet d1
Treatment
Production mixers 0
Degasser
Filters hot
Gas makeup
Heat exchanger
Filters cold
Injection pump
Flare
Grids
Gas grid
Electricity grid
Atmosphere
Heating
Peak boiler
Drive splitter
Heating supply pipe
Heating consumer
Heating return pipe
Drive mixer
Cooling
Absorption chiller
Cooling supply pipe
Cooling consumer
Cooling return pipe
Absorption cooler
Brine
Gas
Hot water
Air
Chilled water
Electricity